# HOMEWORK EXERCISE 3 SOLUTIONS

1) IF WE CONSIDER THE SAMPLE OBTAINED FROM EACH STUDENT, WE COULD USE THE DIFFERENCE OF GRE SCORE AFTER EXPERIMENT − GRE SCORE BEFORE (MINUS)

IN SUCH A CASE, THE NULL HYPOTHESIS, OR STATUS QUO, WOULD BE TO SUGGEST NO CHANGE IN SCORE HAS OCCURRED. "NO CHANGE" IMPLIES $= 0$.

2) $H_a: \mu > 0$ (PRACTICING MEDITATION BOOSTS GRE SCORE, ON AVERAGE THAN IF everyone WERE TO TAKE A NUTRITION CLASS)

3) A TYPE I ERROR OCCURS WHEN WE REJECT $H_0$ WHEN IT IS TRUE

HERE, THIS COULD MEAN THAT IN REALITY, MEDITATION DOES NOT INCREASE GRE SCORE, BUT WE CONCLUDE THAT IT DOES

4) A TYPE II ERROR OCCURS WHEN WE DO NOT REJECT $H_0$, WHEN IN FACT, IT IS FALSE.

HERE, THIS COULD MEAN THAT IN REALITY, MEDITATION DOES INCREASE MEAN GRE SCORE, BUT WE CONCLUDE THAT IT DOESN'T (OR, WE FAIL TO MAKE THAT CONCLUSION).

5) POSSIBLE SOLUTION/RESPONSE: IT SEEMS LIKE MAKING A TYPE I ERROR WOULD BE MORE SERIOUS BECAUSE A USEFUL METHOD FOR INCREASING TEST SCORES WOULD GO UNDISCOVERED.